

Joint angle calculation from optical markers and IMUs attached to the lower body

The purpose of this protocol was to clarify the derivation of joint angles from both inertial measurement units (IMU system) and an optical motion capture (OMC) system as used in the article “Validity of inertial sensor based 3D joint kinematics of static and dynamic sport and physiotherapy specific movements”.

Notation

In this protocol, a rotation from coordinate frame B to frame A was denoted R^{AB} , where $(R^{AB})^T = R^{BA}$ was the inverse rotation. Rotations were parametrized as quaternions; e.g. q^{AB} was the quaternion representation of R^{AB} . The vector B^A denoted the location of the origin of coordinate frame B given in A . The vector p_i^A denoted the i^{th} point in coordinate frame A .

Physical setup

The physical setup is shown in Figure 1. Retroreflective markers were placed on a participant according to the protocol provided in Leardini et al. [1]. The segment coordinate systems (S) were created from the tracked 3D marker positions (NaturalPoint OptiTrack, 12 cameras) according to Cappozzo et al. [2]. For each segment S_i , the 3D positions of the associated markers and joint centers were denoted $p_i^{S_i} \in P^{S_i}$ (Figure 1 “Points in coordinate frame S_i ”). Each IMU was rigidly mounted in a 3D printed casing with four retroreflective markers attached to it (Figure 2). The markers were combined to rigid bodies using the OMC system and the positions B^G and orientations q^{GB} of the associated coordinate systems B with respect to the reference frame G were provided by the OMC system along with the single marker positions. Note, G was defined through the initial OMC system calibration and it was leveled so that z^G opposes gravity. Hence, G denotes both the OMC and the IMU reference frame. IMU measurements were provided in coordinate system I , which was assumed in the physical location of the accelerometer triad. Note, the marker rigid body coordinate systems were constructed so that their origins coincide with the origins of the associated IMU coordinate frames.

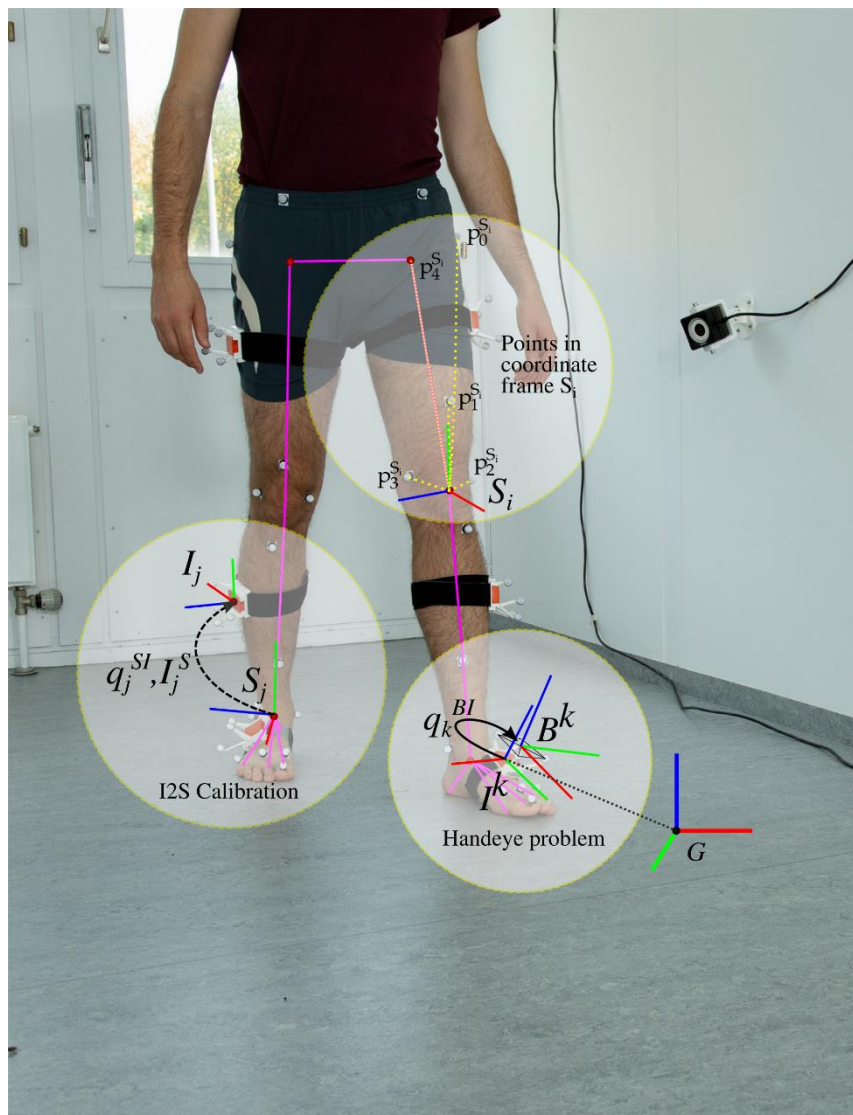


Figure 1: Coordinate systems and biomechanical model.

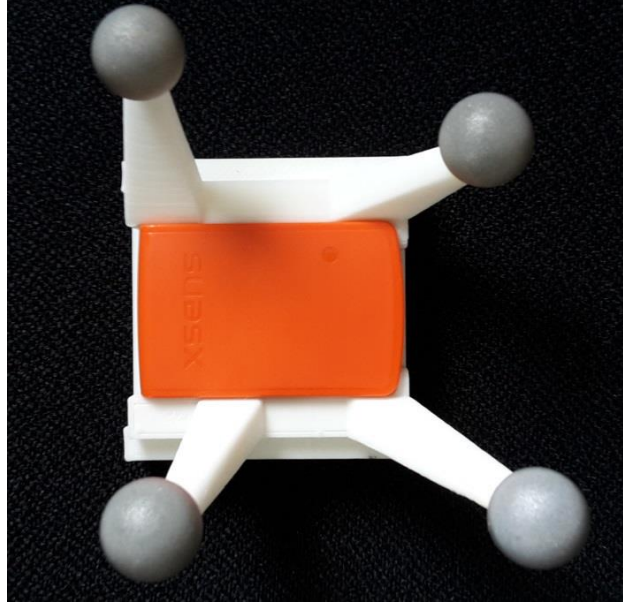


Figure 2: Inertial measurement unit inserted in 3D printed casing with four retroreflective markers

Segment pose estimation from OMC data

Let $S_{i,t}^G, q_{i,t}^{GS}$ be the sought global position and orientation of segment i at time t . This transformation was defined through a set of $n \geq 3$ linearly independent 3D point correspondences, i.e. the marker positions in the local segment coordinate frame $\{p_{1:n}^{S_i}\}$ and their observed positions in $G \{p_{i,j,t}^G\}$ as provided by the OMC system (Figure 1). The complete trajectory of segment i (i.e. the pose of each frame $t = 1..m$) was then obtained by minimizing:

$$\{S_{i,t}^G, q_{i,t}^{GS}\}_{t=1}^m = \underset{\{S_{i,t}^G, q_{i,t}^{GS}\}_{t=1}^m}{\operatorname{argmin}} \sum_{t=1}^m \operatorname{err}(S_{i,t})$$

with:

$$\operatorname{err}(S_{i,t}) = \sum_{j=1}^n \|S_{i,t}^G + R_{i,t}^{GS} p_j^{S_i} - p_{i,j,t}^G\|_2$$

When calculating this for each segment independently, the segments will likely not stay connected at the joints during the estimated movement. To enforce this, joint constraints were added to the optimization problem. As stated above, joint centers were, just like skin markers, defined as points in the local segment coordinate frames. A joint was defined as a four tuple of indices $J = \{i, j, k, l\}$ carrying the information that $p_j^{S_i}$ and $p_l^{S_k}$ should be connected in the global frame G . If $J_p = \{i, j, k, l\}$ is the p^{th} joint in the biomechanical model, the joint center distance at time t was:

$$\operatorname{err}(J_{p,t}) = \|p_{i,j,t}^G - p_{k,l,t}^G\|_2 = \|S_{i,t}^G + R_{i,t}^{GS} p_j^{S_i} - S_{k,t}^G + R_{k,t}^{GS} p_l^{S_k}\|_2$$

The resulting constrained optimization problem for deriving the movement of the complete biomechanical model including s segments connected through k joints at m frames was:

$$\{S_{i,t}^G, q_{i,t}^{GS}\}_{i=1,t=1}^{s,m} = \underset{\{S_{i,t}^G, q_{i,t}^{GS}\}_{i=1,t=1}^{s,m}}{\operatorname{argmin}} \sum_{t=1}^m \left(\sum_{j=1}^s \operatorname{err}(S_{j,t}) + \lambda \sum_{p=1}^k \operatorname{err}(J_{p,t}) \right)$$

where λ was a weight parameter to adjust the influence of the joint constraints. The above described calculations follow the definitions of Visual3D (C-Motion, Inc, Germantown, MD, USA), a widely used software tool for 3D biomechanics research. Choosing $\lambda = 0.01$ showed the best correspondence to segment poses obtained from Visual3D.

Handeye estimation

In this protocol, the handeye problem refers to the problem of calculating the rigid transformations between I and B for each IMU/rigid body pair. This was required for being able to relate the IMU measurements to the OMC rigid body data (Figure 1 “Handeye problem”). It was subsequently used for obtaining the IMU-to-segment (I2S) calibrations from the OMC data as well as for initializing the sensor fusion method for segment pose estimation from the IMU measurements. Since the origins of I and B coincided (as mentioned above), this came down to calculating the orientations q^{BI} for each IMU/rigid body pair. This was done based on a synchronized sequence of angular velocities ω^I measured by the IMUs and angular velocities ω^B derived from the marker rigid body orientations (first derivative), with $\omega^B = q^{BI} \omega^I$, according to the method proposed in Chardonnes et al. [3]. The data sequence was obtained by mounting the casings containing the IMUs on a bar and rotating the bar in all directions in the tracking volume.

I2S Calibration

The IMU-to-segment (I2S) calibration refers to the relative orientation q^{SI} between the IMU and the segment it is attached to (Figure 1 “I2S Calibration”). In order to exclude errors due to pose based or functional calibration methods, in this protocol, the I2S calibration was extracted from a static neutral pose sequence by exploiting the OMC data and the above described handeye orientation. The I2S orientation was:

$$R^{SI} = R^{SG} R^{GB} R^{BI}$$

where R^{SG} was the OMC segment orientation, R^{GB} was the OMC rigid body orientation and R^{BI} was the handeye orientation. Note, the indices are omitted since every segment was assumed to have one IMU attached to it. The I2S position was:

$$I^S = B^S = R^{SG}(B^G - S^G)$$

where B^G was the global rigid body position and S^G was the global segment position. In this protocol, the calibration was assumed to be fix. Thus, OMC based global segment orientations could be either obtained using the skin markers and the optimization based method described above, or they could be deduced from the rigid body orientations using:

$$R^{GS} = R^{GB} R^{BI} R^{IS}$$

Segment pose estimation from IMU data

The sensor fusion method for estimating the segment poses from IMU data assuming known I2S calibrations and known initial IMU poses was completely described in Teufl et al. [4]. As for the I2S calibrations, in this protocol, the initial IMU poses were obtained from the OMC data and the above described handeye transformation.

Joint angle decomposition

A joint orientation was here defined as the relative orientation between the connected segments; e.g. the right hip orientation was:

$$q_{RightHip}^{SS} = q_{Pelvis}^{SG} \odot q_{RightUpperLeg}^{GS}$$

where the global segment orientations were obtained using one of the methods above (OMC skin markers, OMC rigid bodies, IMU). The joint angles were then obtained by decomposing the relative joint orientations according to the coordinate system definitions in Cappozzo et al. [2] and the decomposition order proposed in Visual3D, i.e. flexion (θ_z) followed by abduction (θ_x) followed by internal rotation (θ_y). The Euler angle decomposition is taken from [5].

References

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